OBJECTIVES::

The objective of the paper is to provide fundamental of the numerical simulation technique and its application in geomechanics.

COURSE TOPICS::

Finite element method

Concept, discretization into elements, element types, element stiffness, assemblage and solution. Simulation based on fem.

Finite difference method

Concept, formation of mesh, finite difference patterns, solutions.

Boundary element method

Concept, discretization, solution for isotropic and infinite media.

Solver

Iterative solution scheme for simulations equations

READINGS

TEXTBOOKS::

REFERENCE BOOKS::

1. Tirupathi R. Chandrupatla, Ashok D. Belegundu, Introduction to Finite Elements in Engineering: International Edition, Pearson Education Limited, 2014

2. Christian Duenser, Gernot Beer, and Ian Moffat Smith, The Boundary Element Method with Programming: For Engineers and Scientists, Springer;Softcover reprint of hardcover 1st ed. 2008

3. Olek C Zienkiewicz, Robert L Taylor, The Finite Element Method for Solid and Structural Mechanics, Butterworth-Heinemann, Technology & Engineering